

Homework & Tutorial

Homework: 5.2, 5.5, 5.15, 5.21(a-f, h)

Tutorial Problems: 5.1, 5.3, 5.4, 5.41

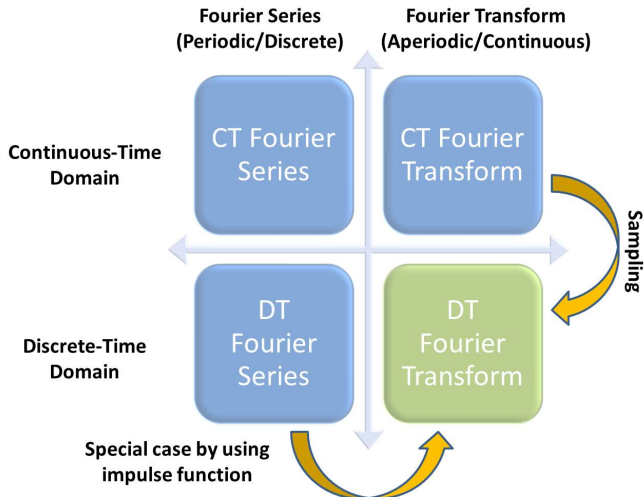
Chapter 5: The Discrete-Time Fourier Transform

Department of Electrical & Electronic Engineering

2022-Fall

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A Big Picture



Outline

- Definition of discrete time Fourier transform (DTFT)
 - ▶ From Fourier series to Fourier transform
 - ▶ Convergence issue
- Periodic signals' DTFT
- Properties of DTFT

Discrete-Time Fourier Transform

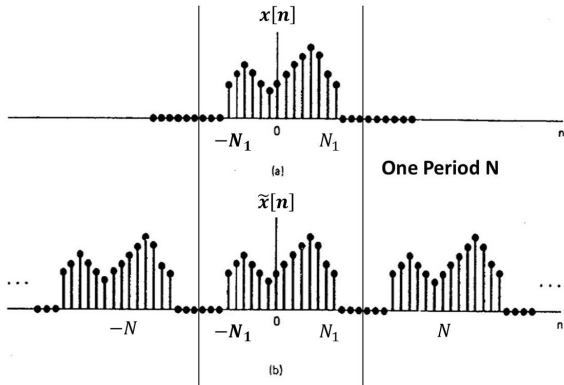
- Discrete-time Fourier series (for periodic signals)

$$\text{Synthesis Equation: } x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$\text{Analysis Equation: } a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

- Aperiodic signals can be treated as periodic signals with period $N \rightarrow \infty$
 - $x[n]$ must be like $\sum_k b_k e^{jk(2\pi/N)n}$ or $\int_{\omega} b(\omega) e^{j\omega n} d\omega$
 - b_k or $b(\omega)$ can be calculated from $x[n]$

Derivation (1/3)



- Original signal: $x[n]$
- Define new periodic signal with period N : $\tilde{x}[n]$, such that

$$\tilde{x}[n] = x[n], \quad n = -N/2, \dots, N/2 - 1$$

- Notice: when $N \rightarrow \infty$, $\tilde{x}[n]$ becomes $x[n]$

Derivation (2/3)

- Look at the Fourier series of $\tilde{x}[n]$:

$$Na_k = \sum_{n=-N/2}^{N/2+1} \tilde{x}[n] e^{-jk(2\pi/N)n} = \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$

- Define $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$, so that $Na_k = X(e^{jk\omega_0})$ ($\omega_0 = 2\pi/N$)
- Look at the synthesis equation:

$$\begin{aligned}\tilde{x}[n] &= \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} = \frac{1}{N} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \\ &= \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \omega_0\end{aligned}$$

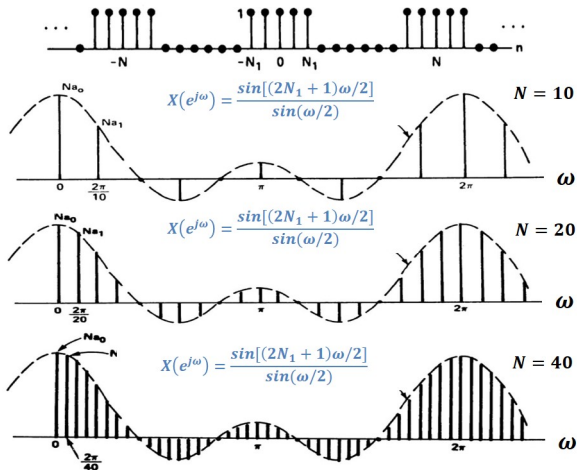
- When $N \rightarrow \infty$, $\omega_0 \rightarrow d\omega$, $\Sigma \rightarrow \int$, $k\omega_0 \rightarrow \omega$, $\tilde{x}[n] \rightarrow x[n]$, thus,

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Navigation icons: back, forward, search, etc.

Example: From Periodic To Aperiodic

$$Na_k = \frac{\sin \frac{2\pi k \left(N_1 + \frac{1}{2}\right)}{N}}{\sin \left(\frac{\pi k}{N}\right)} = X(e^{j\frac{2\pi k}{N}})$$



Larger N , Denser Harmonic Components

Derivation (3/3)

- Therefore, we get the discrete-time Fourier transform pair

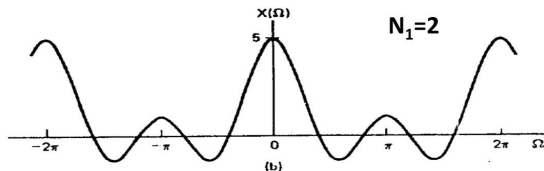
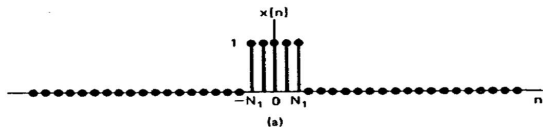
Discrete-Time Fourier Transform

$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- Observations
 - ▶ Continuous spectrum: Similar to CTFT
 - ▶ Periodic with period 2π : Different from CTFT
 - ▶ Low frequency: close to 0 and 2π ; high frequency: close to π

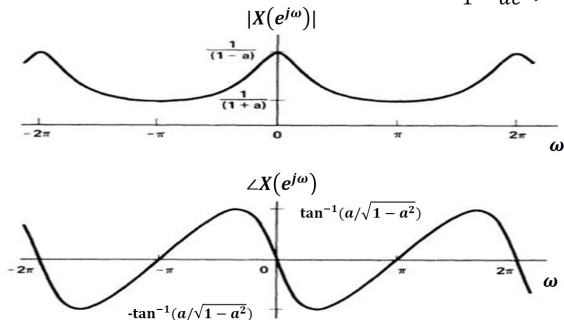
Fourier Transform Examples (1/2)



- $x[n] = 1$ ($n = -N_1, \dots, 0, \dots, N_1$)
- $X(e^{j\omega}) = \frac{\sin(\omega(N_1+1/2))}{\sin(\omega/2)}$
- Width of $x[n]$: $W_t = 2N_1 + 1$; width of $X(e^{j\omega})$: $W_f = \frac{4\pi}{2N_1+1}$
- $W_t \times W_f = 4\pi$, which is a constant

Fourier Transform Examples (2/2)

$$x[n] = a^n u[n] \quad 0 < a < 1 \quad \Leftrightarrow \quad X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$



- See textbook, Example 5.1
- What's the shape of magnitude when $a \rightarrow 1$ or $a \rightarrow 0$?

Convergence Issue of Analysis Equation

Sufficient Condition of Convergence

The analysis equation $X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n]e^{-j\omega n}$ will converge either if $x[n]$ is absolutely summable or if the sequence has finite energy, thus,

$$\sum_{-\infty}^{\infty} |x[n]| < \infty \text{ or } \sum_{-\infty}^{\infty} |x[n]|^2 < \infty$$

• Do the following signals have Fourier transform:

- ▶ $a^n u[n]$ ($0 < a < 1$)
- ▶ $\delta[n]$
- ▶ $u[n]$
- ▶ $e^{j\frac{2}{5}\pi n}$, $\cos(\frac{2}{5}\pi n)$
- ▶ $a^n u[n]$ ($a > 1$)

Can Periodic Signals Have DTFT?

- Definition of DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- Justification of divergence:

- ▶ Let $\omega = 2k\pi$, we have $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]$
- ▶ Since $x[n]$ is periodic, the summation $\sum_{n=-\infty}^{\infty} x[n]$ will never converge unless $x[n] = 0$

- **Conclusion:** Most of periodic signals do NOT have DTFT according to the definition
- However, it's of significant engineering importance to extend Fourier transform to periodic signals

DTFT with Periodic Signals (1/2)

Fourier Transform of $e^{j\omega n}$

The following transform pair is actually NOT rigorously defined:

$$x[n] = e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)$$

- Synthesis:

$$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{2\pi} \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l) e^{j\omega n} d\omega = e^{j\omega_0 n}$$

- Analysis:

$$\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} e^{j(\omega_0 - \omega)n} \quad \text{converge??}$$

Remark

- We just **believe** the following equation is true:

$$\sum_{n=-\infty}^{\infty} e^{j\omega n} = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - 2\pi l)$$

- When $\omega = 2k\pi$,

$$\sum_{n=-\infty}^{\infty} 1 = 2\pi\delta(0)$$

DTFT with Periodic Signals (2/2)

- According to the Fourier series, for a periodic signal with period N :

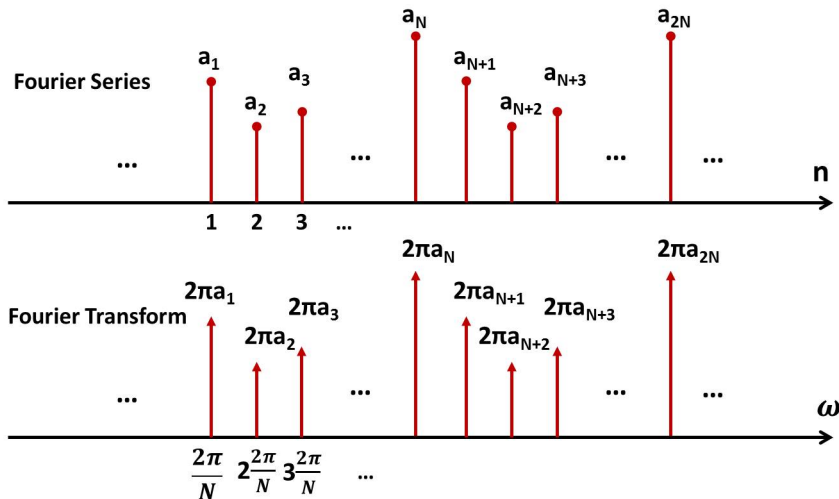
$$x[n] = a_0 + a_1 e^{j(2\pi/N)n} + \dots + a_k e^{jk(2\pi/N)n} + \dots + a_{N-1} e^{j(N-1)(2\pi/N)n}$$

- $e^{jk(2\pi/N)n} \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l)$
- Then, due to the linearity of Fourier transform

$$\begin{aligned} \mathcal{F}\{x[n]\} &= \sum_{k=0}^{N-1} a_k \mathcal{F}\{e^{jk(2\pi/N)n}\} = \sum_{k=0}^{N-1} a_k \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \\ &= \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} a_k 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \end{aligned}$$

- Fourier transform of a periodic signal is a periodic sequence of impulses
 - What's the period? How many impulses within one period?

Fourier Series v.s. Fourier Transform



Example: Discrete-Time Impulse Chain

- What's the Fourier transform of $x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$?
- First of all, we calculate the Fourier series:

$$\begin{aligned}a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{k=-\infty}^{+\infty} \delta[n - kN] e^{-jk \frac{2\pi}{N} n} \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N}\end{aligned}$$

- Then, we have

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} \frac{2\pi}{N} \delta(\omega - k(2\pi/N) - 2\pi l) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{N})$$

- Time domain period $\times$ Frequency domain period = ?



Periodicity, Linearity and Shifting

- Periodicity

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

- ▶ How about CTFT? Why?

- Linearity

$$ax_1[n] + bx_2[n] \longleftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$$

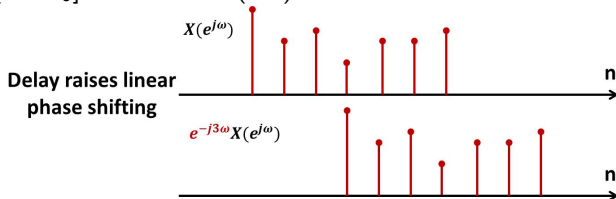
- Time Shifting and Frequency Shifting

$$\begin{aligned}x[n - n_0] &\longleftrightarrow e^{-j\omega n_0} X(e^{j\omega}) \\e^{j\omega_0 n} x[n] &\longleftrightarrow X(e^{j(\omega - \omega_0)})\end{aligned}$$

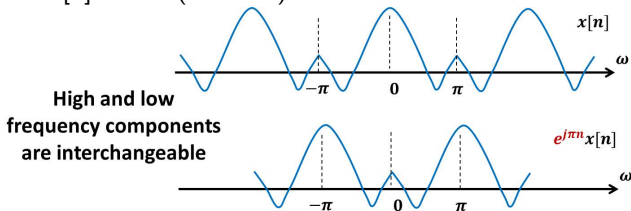
- ▶ What's the physical meaning?

Illustration on Shifting

- $x[n - n_0] \longleftrightarrow e^{-j\omega n_0} X(e^{j\omega})$



- $e^{j\omega_0 n} x[n] \longleftrightarrow X(e^{j(\omega - \omega_0)})$

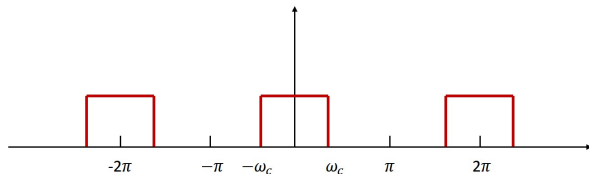


Example: Shifting

- Similar to CT, the DTFT of **impulse response** of a DT LTI system is **frequency response**.

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]e^{-j\omega n}$$

- Given an ideal LPF with the following frequency response



- Suppose its impulse response is $h_{lpf}[n]$, please predict the behavior of another LTI system with impulse response $(-1)^n h_{lpf}[n]$.

Conjugation, Differencing and Accumulation

- Conjugation

$$x^*[n] \longleftrightarrow X^*(e^{-j\omega})$$

- ▶ $X(e^{j\omega}) = X^*(e^{-j\omega}) \Leftrightarrow x[n]$ is real
- ▶ If $x[n]$ is real, then $\Re\{X(e^{j\omega})\}$ is even, $\Im\{X(e^{j\omega})\}$ is odd

- Differencing

$$x[n] - x[n-1] \longleftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$$

- ▶ High-pass or low-pass?

- Accumulation

$$\sum_{m=-\infty}^n x[m] \longleftrightarrow \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

- ▶ How to derive it via differencing?

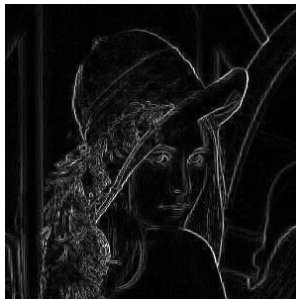
Problem 5.37

Problem

Let $X(e^{j\omega})$ be the Fourier transform of $x[n]$. Derive expressions in terms of $X(e^{j\omega})$ for the Fourier transforms of the following signals.

- $\text{Re}\{x[n]\}$
- $x^*[-n]$
- $\text{Ev}\{x[n]\}$

Effect of Differencing



- $J(i,j) = |M(i,j) - M(i+1,j+1)| + |M(i+1,j) - M(i,j+1)|$

Fourier Transform of $u[n]$

- How to derive the Fourier transform of $u[n]$?
- Option 1: From definition of Fourier transform

$$\mathcal{F}\{u[n]\} = \sum_{n=-\infty}^{+\infty} u[n]e^{-j\omega n} = \sum_{n=0}^{+\infty} e^{-j\omega n} \quad \text{Converge??}$$

- Option 2: Since $u[n] = \sum_{m=-\infty}^n \delta[m]$, according to the property of accumulation

$$\mathcal{F}\{u[n]\} = \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

- Observation: Fourier transform of $u[n]$ does not exist according to the definition; however, it can be expressed in terms of $\delta(\cdot)$

Time Reversal and Expansion

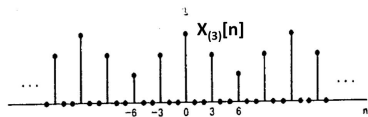
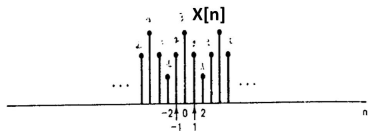
- Time Reversal

$$x[-n] \longleftrightarrow X(e^{-j\omega})$$

- Time Expansion

► Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega}) \quad (1)$$



Differentiation and Parseval

- Differentiation in Frequency

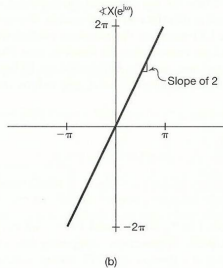
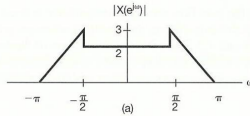
$$nx[n] \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

- Parseval's Relation

$$\sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

- ▶ Energy density spectrum — $\frac{|X(e^{j\omega})|^2}{2\pi}$

Example



- See textbook, Example 5.10
- Spectrum within $[-\pi, \pi]$
- Is it periodic, real, even, of finite energy?